

International Summer School 2026

Information package for participants

Hello everyone,

It's almost time for our International Summer School! Here you'll find a concise summary of the most important information about the venue, accommodations, online program, activities in Karlsruhe, and the WhatsApp group.

URGENT: Please complete this form no later than 29

August: <https://forms.cloud.microsoft/e/LGSJ05QaES>

1. Venue

On-Campus-Week: Hochschule Karlsruhe (HKA), Karlsruhe, Deutschland

Duration: 6.–11. September 2026

The program takes place at the HKA, as well as, in part, during field trips and technical tours in and around Karlsruhe.

Site Plan:



2. Whatsapp group

Please join until 06.09.2026.

NBi3s 2026

WhatsApp-Gruppe



Link: <https://chat.whatsapp.com/FUALJFEzuir9gABJLOGCUG?s=qt&p=i&mlu=4&amv=0>

3. Online-Program

The online sessions are part of the Summer School and serve as preparation for the week we'll spend together in Karlsruhe.

The online lectures will be conducted by the University of Oklahoma (USA).

The link to join the online sessions:

<https://okstate-edu.zoom.us/j/98877639461?pwd=IQzbyQ2gOrFHFzK7avJQcE55u41TvC.1>

If you are unable to attend one of the online sessions, please let us know briefly. The presentation slides will be shared with you afterward.

Day	Date	Dur atio n	Start time				End Time				Topic (Presenter)
		hrs	Okla hom a	Ger man y	Kore a	Mal aysi a	Okla hom a	Ger man y	Kore a	Mala ysia	
Mond ay	2026-08-31	1	7:00 AM	2:00 PM	9:00 PM	8:00 PM	8:00 AM	3:00 PM	10:0 0 PM	9:00 PM	Introduction of all participants and lecturers (all)
		2,5	8:00 AM	3:00 PM	10:0 0 PM	9:00 PM	10:30 AM	5:30 PM	12:3 0 AM	11:3 0 PM	Biomimicry, Biophilic Design and Industrial Ecology (Aditya)
Tuesd ay	2026-09-01	2,5	8:00 AM	3:00 PM	10:0 0 PM	9:00 PM	10:30 AM	5:30 PM	12:3 0 AM	11:3 0 PM	Sustainability in Building Conditioning Systems (Christian)
Wedn esday	2026-09-02	2,5	8:00 AM	3:00 PM	10:0 0 PM	9:00 PM	10:30 AM	5:30 PM	12:3 0 AM	11:3 0 PM	Design of Low-Impact Occupant Centric Buildings (Khaled)
Thurs day	2026-09-03	2,2	8:00 AM	3:00 PM	10:0 0 PM	9:00 PM	10:30 AM	5:30 PM	12:3 0 AM	11:3 0 PM	Making Existing Buildings Great Again! (Heba)
Thurs day	2026-09-03	1,0	8:00 AM	3:00 PM	10:0 0 PM	9:00 PM	9:00 AM	4:00 PM	11:0 0 PM	10:0 0 PM	IMPORTANT INFORMATION for the on-campus week

4. On site program – HKA Karlsruhe

Sund ay		06.09.20 26		Climat e Arena Climate Change		
Time	Responsib le	Affiliation	Theme	Subject	Room	Remarks
12:00 - 13:00	Jiyan Akyol	HKA	Bus drive	Transport from Youth Hostel Karlsruhe to Klima Arena Sinsheim		BUS DEPARTURE 12:00
13:00 - 17:00	Michael Kauffeld	HKA	Climate Change	Climate Cahnge	Klima Arena	https://klima-arena.de/
Around 15:00	Coffee and cake			Coffee Break	Klima Arena	
17:00 - 18:00	Michael Kauffeld	HKA	Bus Drive	Return to Karlsruhe		
18:30 - 20:00	Michael Kauffeld	HKA	Dinner	Joint dinner at Restaurant "Treffpunkt1721", Adenauerring 32, 76131 Karlsruhe		to be confirmed

Mond ay		07.09.20 26		Sustai nabilit y Sustainability		
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Time	Responsible	Affiliation	Theme	Subject	Room	Remarks
08:00 - 08:30	Jiyan Akyol	HKA	Registration	Group work assignment / enrolment	Foyer in front of F316	Croissant, sandwich, coffee, tea, juice
08:30 - 08:40	Franz Quint + Michael Kauffeld	HKA	Welcome	Welcome on behalf of HKA + sustainability at HKA	F316	
08:40 - 08:45	Andrea Bühler	City Karlsruhe	Welcome	Welcome from city of Karlsruhe, Sustainable projects in Karlsruhe	F316	
08:45 - 09:30	Andrea Bühler + Felix Zechiel	City + Technology Region Karlsruhe	Lecture	Sustainable project in and around Karlsruhe	F316	
09:30 - 09:50				Coffee Break		
09:50 - 10:35	Henrik Nordborg	OST	Lecture	Global Climate Compensation	F316	
10:35 - 11:20	Michael Kauffeld	HKA	Lecture	PFAS - the next big global challenge	F316	
11:20 - 11:30				Coffee Break		
11:30 - 12:15	Philipp Petry	Impact Strategies GmbH		Sustainable AI	F316	
12:15 - 13:00	Shaina Jain	University of Portsmouth London	Lecture	Reimagining Agriculture: Sustainable, Circular, and Resilient Food Systems	F316	
13:00 - 14:00	Mensa			Lunch break		
14:00 - 14:45	Regina Kratt	RawTwist GmbH	Workshop	Sustainable, micronutrient-rich diet and ice cream	F316	ice cream workshop
14:45 - 17:00	Jiyan Akyol	City Tour	Team building	Treasure hunt through Karlsruhe in groups à 6 students	City of Karlsruhe	
18:00 - 21:00	Jiyan Akyol	HKA	Get together	Lab visits IKKU + Barbecue at Fireplace at HKA + HKA micro brewery	Li015	

Tues day	08.09.2026	Buildings	Refrigeration, Heat Pumps, Thermal Storage			
Time	Responsible	Affiliation	Theme	Subject	Room	Remarks
08:00 - 08:45	All	HKA	Group work	work in the 10 groups	F316	
08:45 - 09:30	Constanze Bongs	HKA	Lecture	Heat pumps as gamechanger in global decarbonisation	F316	
09:30 - 09:50				Coffee Break		

09:50 - 10:35	Michael Kauffeld	HKA	Lecture	Sustainable Refrigeration, Air Conditioning and Heat Pump Technology	F316	
10:35 - 11:20	MD Mahfizur Rahman	OST	Lecture	SophiA Systems	F316	
11:20 - 11:30				Break		
11:30 - 12:15	Tobias Hirsch	Everllence	Lecture	Mega size heat pumps	F316	
12:15 - 13:00	Leon Brendel	OST	Lecture	Industrial Heat Pumps	F316	
13:00 - 13:30	Michael Kauffeld + Jiyan Akyol	HKA	Walk + Talk	Walk through the castle gardens to Mensa Adeneuerring 7	outdoor	
13:30 - 14:15	Studierend enwerk Karlsruhe			Lunch break at KIT Mensa, Vielmetter Hall	Mensa	Coupons for everyone, sitting in Annex close to update
14:30 - 16:00	Michael Kauffeld + Constanze Bongs	HKA	Technical visit	Visit refrigeration system at Mensa KIT / RoofKIT	Adeneuerring 7	two groups of 30 students each alternate between Mensa and RoofKIT
14:30 - 16:00	Constanze Bongs + Michael Kauffeld	HKA	Technical visit	Visit sustainable building RoofKIT / refrigeration system at Mensa	KIT Campus South	two groups of 30 students each alternate between Mensa and RoofKIT
16:00 - 16:30				Drive to LTC		
16:30 - 18:00	Philipp Linder + someone from Secon	LTC	Technical visit	Energy plus building and heat pumps	LTC	LTC – Linder Technology Campus Wilhelm-Schickard-Straße 9 76131 Karlsruhe
18:00 - 19:30	all		Dinner	Dinner in Cantina of LTC	LTC	
from 19:30	all			walk, NextBike, El-Scooter or tram back to Youth Hostel		

Wednesday 09.09.2026 Water and PFAS						
Time	Responsible	Affiliation	Theme	Subject	Room	Remarks
08:00 - 11:30	KNW (Competence Center for Sustainability Workshops)	HKA	Workshop	1. The Circular Economy Collage (F109) 2. The Digital Collage (F316) 3. The Mobility Fresco (IO Tech Café) 4. The Planetary Boundaries Fresco (F109) 5. The Biodiversity Collage (F109)	F108, F110, F109, F316 and IO Tech Café	
when ever	Jiyan Akyol	HKA	Coffee Break	6. The New Narratives Collage (F110)		

time permits				7. The 2 tonnes / Hability workshop (F110) 8. Our Sustainable Lives (F108) 9. The Supply Chain Game (F316)		
11:45 - 12:30	Ursula Schön-Herrmann	DAA	Workshop	How to achieve and maintain employability in an ever faster changing environment	F316	
12:30 - 13:30	Michael Kauffeld	HKA	Drive to Rastatt	Lunch break in the bus	Bus	
13:40 - 15:10	Michael Kauffeld		Excursion	Rastatt PFAS		https://www.stadtwerke-baden-baden.de/de/aktuelles/meldungen/Trinkwasser-Special.php
14:30 - 15:10	Michael Kauffeld		Excursion	Water works Sandweiler	Baden Baden	30 students at Wasserwerk Sandweiler; 30 students at Landseehof
15:10 - 15:20	Bus			Exchange groups		
15:20 - 15:50	Landseehof Haueneberstein			Coffee Break		
15:50 - 16:00	Bus			Exchange groups		
16:00 - 16:45	Michael Kauffeld		Excursion	Rastatt PFAS	Baden Baden	https://www.landseehof.de/
16:45 - 17:30	Michael Kauffeld		Excursion	Landseehof Haueneberstein	Baden Baden	30 students at Wasserwerk Sandweiler; 30 students at Landseehof
17:30 - 18:15			Relax	Drive by coach bus from Walachischer Kessel		
18:15 - 23:00	Cristian Cirstea	Walachischer Kessel	Dinner	Romanian evening at Walachischer Kessel incl. Dinner and unikat concert	Karlsruhe	

Thurs day	10.09.2026	Energy Renewable Energy				
Time	Responsible	Affiliation	Theme	Subject	Room	Remarks
08:00 - 08:45	All	HKA	Group work	work in the groups	F316	
08:45 - 09:30	Mihaela Dudita-Kauffeld	OST	Lecture	Solar Energy - the eternal energy source	F316	
09:30 - 09:50				Coffee Break		

09:50 - 10:35	Laura Comella + Jonathan Larochelle	HKA	Lecture	The Environmental Cost of Saving Paper: A Case Study of Electronic Shelf Labels	F316	
10:35 - 11:20	Maurice Kettner	HKA	Lecture	Ammonia as hydrogen carrier	F316	
11:20 - 11:30				Break		
11:30 - 12:15	Daniel Silsbee	Advansor	Lecture	Sustainability in heating and cooling with CO2	F316	
12:15 - 13:00	Dustin Ochmann	HKA	Lecture	Lilon batteries	F316	PhD of Olivier Schecker
13:00 - 14:00	Michael Kauffeld	HKA		Lunch break in bus	Bus	
14:00 - 15:30	Michael Kauffeld	HKA	Excursion	Visit KIT Campus North Energy Lab + atmospheric research	KIT Campus North	
15:30 - 16:00				Coffee break	KIT Campus North	
16:00 - 16:45	Michael Kauffeld	HKA	Excursion	Visit KIT Campus North Energy Lab + atmospheric research	KIT Campus North	
16:45 - 17:00	Michael Kauffeld	HKA		Drive to TWK		
17:00 - 18:00	Michael Kauffeld	HKA	Group Work	Sustainable Building + Refrigeration Technology	TWK	
18:00 - 18:30	all		drive back to Restaurant	Bus		
18:30 - 20:30	all		Dinner	Dinner at Café Bleu		

Friday 11.09.2026		Climate Disaster Global Warming			Please NOT Different time slots	
Time	Responsible	Affiliation	Theme	Subject	Room	Remarks
08:00 - 08:25	Group 1		Workshop	1. Designing a Net-Zero Energy Building	F316	done by the students
08:25 - 08:50	Group 2		Workshop	2. Low-Impact Occupant-Centric Buildings	F316	done by the students
08:50 - 09:15	Group 3		Workshop	3. Water-Energy Nexus in Hospitals and Communities	F316	done by the students
09:15 - 09:30				Coffee break		

09:30 - 09:55	Group 4		Worksh op	4. Circular Economy for Building Materials	F316	done by the students
09:55 - 10:30	Group 5		Worksh op	5. Solar-Powered Cooling for Food and Medicine	F316	done by the students
10:30 - 10:55	Group 6		Worksh op	6. Urban Agriculture and Circular Food Systems	F316	done by the students
10:55 - 11:20	Group 7		Worksh op	7. The role of bees in a sustainable future	F316	done by the students
11:20 - 11:30				Coffee break		
11:30 - 11:55	Group 8		Worksh op	8. Sustainable Mobility for Campuses and Cities	F316	done by the students
11:55 - 12:20	Group 9		Worksh op	9. Community Engagement and Behavioral Change for Sustainability	F316	done by the students
12:20 - 12:45	Group 10 or 11		Worksh op	10. Resilient Cities in the Face of Climate Change OR Sustainable digital transformation	F316	done by the students
13:00 - 14:00				Lunch break	Mens a	
14:00 - 15:00	Olivier Schecker, Maurice Kettner	HKA	Worksh op	Build up the Climate Fresk in a team		4-8 participants per team
15:00 - 15:45	Olivier Schecker, Maurice Kettner	HKA	Worksh op	Build up the Climate Fresk in a team		4-8 participants per team
15:45 - 16:00				Coffee break		Outdoor, if the weather is nice
16:00 - 16:45	Olivier Schecker, Maurice Kettner	HKA	Worksh op	Summarize what you have learned and choose a title for your fresk		4-8 participants per team, OHNE Publikum
16:45 - 17:30	Olivier Schecker, Maurice Kettner	HKA	Worksh op	See where we stand and share feelings, thoughts, ideas and actions to create a better future		4-8 participants per team, OHNE Publikum
17:30 - 18:30	Michael Kauffeld	HKA	Closing	Hand out Certificates + get feedback from students		Wrap up - feedback on summer school
18:30 - 19:00	Michael Kauffeld	HKA		Walk and Talk to restaurant		
19:00 - 20:30	all			Dinner at Youth hostel OR HKA Cafeteria (if the cafeteria is open)		
20:30 - 23:00	all	Castle	Fun	Schlosslichtspiele		https://www.schlosslichtspiele.info/en/

Group work	Group 1	1. Designing a Net-Zero Energy Building	Concept for a small-scale building that achieves net-zero energy through passive design, renewables, and efficient systems
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week

Group 2	2. Low-Impact Occupant-Centric Buildings	Building solutions that enhance indoor comfort (thermal, acoustic, visual) while minimizing energy and resource use, inspired by biophilic design
Group 3	3. Water-Energy Nexus in Hospitals and Communities	Integrated systems (solar desalination, rainwater harvesting, greywater reuse) for sustainable water supply with minimal energy use
Group 4	4. Circular Economy for Building Materials	Strategies to reuse, recycle, or replace high-carbon materials (e.g., cement, plastics, insulation) in construction
Group 5	5. Solar-Powered Cooling for Food and Medicine	Scalable, low-cost cooling solutions for reducing food waste and ensuring medical cold chains in underserved regions
Group 6	6. Urban Agriculture and Circular Food Systems	Rooftop or vertical farming concepts powered by renewables, with composting and water recycling loops or other ideas
Group 7	7. The role of bees in a sustainable future	Is honey sustainable? The role of bees in a sustainable future (exploring pollination, biodiversity, food security, and ethical honey production)
Group 8	8. Sustainable Mobility for Campuses and Cities	Integrated mobility plan (walking, cycling, electric micro-mobility, and public transport) for sustainable districts or campuses
Group 9	9. Community Engagement and Behavioral Change for Sustainability	Campaign or toolkit to promote sustainable lifestyles (energy savings, waste reduction, water conservation) at community level
Group 10	10. Resilient Cities in the Face of Climate Change	Adaptation strategies for cities exposed to heatwaves, flooding, or water scarcity, using green infrastructure and sustainable urban design
Group 11	11. Sustainable digital transformation	Digital tools like AI, smart grids, and digital twins can boost efficiency, reduce resource consumption, and enable circular economy models

5. Accomodation

DO'S

- Only use deodorant and hairspray in bathrooms.
- Charge batteries and electrical devices under supervision. When leaving your room, unplug all devices from the power outlet! (Multi-plug adapters are available at reception.)
- Smoke in the designated smoking area (eco tip: dispose of your cigarettes in the ashtray afterwards!).

DONT'S

- The use of electrical appliances for cooking, such as kettles, rice cookers, etc., in the youth hostel.
- Bringing food and drinks—especially alcohol—into the youth hostel.
- Smoking or vaping in the youth hostel (cleaning fee of at least EUR 60.00).
- Bringing and consuming drugs or intoxicants (including cannabis).
- Taking food from the dining room (except for purchased lunches to go).

Costs for damage or loss incurred:

- Lost key: EUR 60.00
- Beds not made with the bed linen provided: EUR 50.00
- Graffiti on walls or beds: EUR 60.00
- Misuse of emergency exits: EUR 100.00
- Smoking in the room / tampering with smoke detectors: EUR 100.00
- Causing the fire department to be called due to smoking, using deodorant or hairspray, using an electric kettle, or similar: EUR 1,750.00

WELCOME TO THE JUGENDHERBERGE KARLSRUHE



Opening hours, quiet hours, and meal times:

Breakfast:	6:30 a.m. - 10:00 a.m.	Check-In:	4:00 p.m. - 9:00 p.m.
Lunch:	12:00 p.m. - 1:00 p.m.	Check-Out:	8:00 a.m. - 10:00 a.m.
Dinner:	5:30 p.m. - 7:00 p.m.		
Quiet hours:	10:00 p.m. - 6:00 a.m.		

Moltkestraße 24 76133 Karlsruhe
Phone: 0721 / 282 48 Mail: jh-karlsruhe@jugendherberge.de



YOUR STAY

- Please use **OUR bed linen** to make your bed and cover your pillow and duvet. For hygiene reasons, you are not permitted to use your own bed linen.
- **Quiet hours** are between 10:00 p.m. and 6:00 a.m. Please respect the quiet hours and behave accordingly so that all guests can feel comfortable with us.
- **Deodorant and hairspray** may only be used in the bathrooms, as the mist can trigger the smoke detectors. Any costs incurred by the fire department will be borne by the person responsible.
- Please wipe down your **table in the dining room** after use and take the used dishes to the dishwashing station. Group leaders should designate a person responsible for each meal.
- If you would like to take breakfast with you, you are welcome to purchase a sandwich (EUR 2.50) or a **packed lunch** (EUR 7.00) to put together yourself. The packed lunch includes 2 sandwiches, a drink, a sweet, and a fruit.
- Please vacate your room by 10:00 a.m. on the day of **check-out** and note the following points:
 - Remove all bed linen and place it in the designated containers. Please fold the duvet and pillow neatly.
 - Return any towels and baby bedding you may have borrowed to the reception desk.
 - Put the windows in the tilt position and turn the heater to level 2.
 - Leave the room swept clean. Brooms are located in the hallways or in the stairwell.
 - Please take your trash to the nearest trash station and separate it into recyclables, paper, and residual waste. Groups should take their trash to the trash containers next to the building.
 - For groups, the reception desk will also conduct a final inspection of the rooms. Please allow enough time for this.

GOOD TO KNOW!

- If you need a **baby cot** or a **safety guard**, we will be happy to provide one for you.
- **Towels** and **toiletries** (toothbrush set, shower gel, etc.) and **earplugs** are available for purchase at the reception.
- You can borrow **multiple plug adapters** at the reception desk.
- There is free **WiFi** in the common areas. All you have to do is connect; no password is required.
- You can buy **drinks, snacks, and ice cream** from the vending machines in the entrance area and at the reception desk.
- **Cards and board games** are available at the reception desk, as well as **balls, table tennis bats**, and other sports equipment.
- The **luggage room, chill room, washing machine** and dryer (for a fee) are located in the basement. The **bicycle storage** room is only accessible from outside and during reception opening hours.
- The Jugendherberge Karlsruhe has a comprehensive **safety concept**. If you feel uncomfortable or notice anything that may require our attention, please contact our team.
- You may encounter **marbled tree bugs** in your room. These are completely harmless and are just looking for a warm place to stay. We do our best to escort them back outside, but sometimes they are simply faster than us.



Emergencies

In urgent matters or emergencies, you can contact our security service **BIG** (tel. 0711 97461028 – call charges apply).

If you need to make an emergency call (110/112), you can use the **emergency telephone to the left of the reception desk**.

We look forward to an exciting and international summer school with you! 🌞🌍

Hochschule Karlsruhe
University of
Applied Sciences

Institut für
**Kälte-, Klima- und
Umwelttechnik**



NOBLE BANADDA INTERNATIONAL
SUSTAINABILITY SUMMER SCHOOL

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